Forecasting Wi-Fi channel occupancy from AERPAW spectrum recordings, including under simulated sparse monitoring: an exploratory deep dive

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Summary. Using public AERPAW sweeps from an urban node (CC1, 7.0 days) and a rural node (LW1, 4.75 days, February 2022), I forecast the *share of sweeps detected busy* in the 2437 MHz band (activity consistent with Wi-Fi channel 6; nothing is decoded) for the 15-minute window starting 15 minutes ahead. (1) **With complete input, a two-hour average is very hard to beat**: six families of learned models were tried and none was reliably better. (2) **These recordings could rule out large gains, not small ones**: a true 5 % improvement would be confirmed on all three recordings only 3.5 % of the time under optimistic assumptions. (3) **With about one sweep in 100 visible, learned models clearly beat plain history but not a simple baseline**: a pre-declared test returned an *honest negative*. All results are exploratory on repeatedly inspected data; the independent test (recording CC2) has not been run.

1 Question, data and scope

How well do recent measurements, and models trained on them, forecast near-future channel activity, and what changes when measurements are missing or late? "Busy" means detected power above a threshold at one sensor: not transmitter activity, decoded traffic or permission to transmit.

Data are the AERPAW fixed-node sub-6 GHz sweeps (Dryad 10.5061/dryad.hmgqnk9zn, CC0), cached in 5 MHz blocks; sweeps arrive about every 19 s (CC1) and 30 s (LW1). A sweep is busy when 20 MHz band power exceeds the noise floor, the 5th percentile of the recording's *first 24 hours*, by 3 dB. The target is the busy share of a 15-minute window. A NIST occupancy release was considered as a second dataset but its file server was unreachable when tried, so it was not used. **Sparsity is therefore simulated**: sweeps are hidden from the forecaster while labels come from the complete stream.

2 Method and discipline

Comparators: the busy share over the last two hours, tuned moving and decayed histories, the long-run average, and history pulled toward that average. Learned models: ridge, gradient boosting, residual versions, a history mixture, local adaptive corrections, state-space corrections and a five-minute count model. Splits are chronological with embargoes; features use only data available at forecast time. Uncertainty is a block bootstrap, because neighbouring windows are alike. An append-only registry records every declared attempt, including failures, and the final test (CC2) is frozen but unrun. Figures and numbers are traceable to saved files by an automated claim check.

3 With complete input: history holds

Attempt	Outcome
Models chosen on earlier data (ridge, boosting)	8.7 % (urban) and 5.5 % (rural) more error than selected history
Learned mixture of eight history forecasts	1.35 % / 6.80 % less error than selected history, but 7.63 % / 1.87 % more than the fixed two-hour average
Shared training, finer inputs	1.99 % / 4.36 % more error than fixed history; 160.45 % more on a reserved later rural session
Local adaptation (145 declared settings)	best gain +0.30 % on first-session rural data; none met the 5 %-on-every-recording rule
State-space residual corrections	urban error up 0.42 % and 0.22 %
Five-minute activity counts	first-session rural error up 22.68 % against fixed history

Table 1. Squared-error differences on common windows; positive favours the learned model. Urban CC1, rural LW1.

Activity persists (a window's busy share correlates 0.75 with the next one at the urban site), the busy definition matters (rural prevalence falls from 30.15 % to 0.70 % when the threshold moves from +2 to +4 dB), and a few windows dominate error: the worst tenth carry 58 % of urban squared error.

4 What could these recordings have shown?

A power analysis from the saved forecasts estimates the smallest true gain each comparison could reliably detect. For a typical candidate it is 21.7 % (urban), 30.8 % (first rural session) and 17.7 % (later rural session). A true 5 % gain would be confirmed on all three recordings 3.5 % of the time (optimistic) or 0.3 % (typical); roughly 86 independent urban days would be needed. The negative results of Section 3 therefore rule out large learned gains, not small ones. Pooling many frequency bands does not rescue this: errors in bands recorded together move together (correlation 0.17–0.28), so the uncertainty floor barely falls. That plan remains a draft.

5 When most measurements are missing

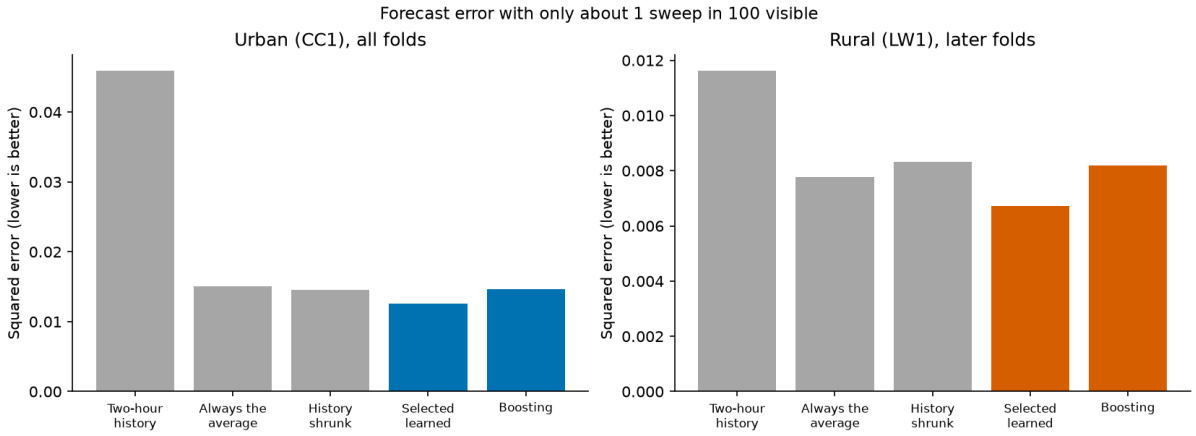


Figure 1: Squared error with about one sweep in 100 visible (lower is better). Plain history collapses; pulling it toward the average removes most of the damage; the learned model is only slightly lower (urban) or no better (rural).

Earlier exploratory runs suggested large learned gains at 99 % dropout, but against plain history and for models trained on complete input. A study with its decision rule written first trained and selected ridge, boosting and residual boosting for the *same* degraded input and compared them with the strongest simple baseline, chosen on earlier data. Urban squared error: plain history 0.0459, strongest simple baseline 0.0146, matched learned model 0.0126 (72.5 % better than plain history, 13.3 % better than the baseline, one-sided 95 % lower bound −33.5 %). Rural later folds: −1.6 % (lower bound −42.0 %). The rule demanded a lower bound of at least 20 % at both sites, so the label is **honest negative**. Training for the degraded view did help the learned model (error 0.0374 became 0.0126). Prediction ranges were not calibrated: the urban 80 % range captured 65.9 % of outcomes. The decision rule's power proxy was amended once, before any matched model was fitted, after its original form triggered; both versions are on record.

6 Limits

Three recordings at two sites in one month; simulated sparsity; roughly 5–14 independent eight-hour blocks per comparison; repeated inspection of all data, so every result is exploratory; uncertainty ranges do not account for the full history of searches. Rural uncertainty is especially weak. Labels measure detected energy only. **CC2, the one independent recording, has not been run.**

7 Reproducibility and status

Code, protocols, saved predictions, the registry and a claim-to-evidence table are in the repository; commands are in `docs/reproduction.md`. Status labels: *exploratory* (all results), *declared in the registry* (not externally timestamped), *frozen and pending* (CC2), *draft* (multi-band). Raw data are not redistributed.